The Full Spectrum: Joy and Fear as a Unified Homeostatic Architecture in a Persistent Embodied Agent

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Abstract

Most approaches to agent safety model threat detection: what is dangerous? This paper argues that framing is half finished. A robust homeostatic architecture must also ask: what makes this okay? What reduces uncertainty? What signals safety? And beyond that: what produces joy? What earns approach rather than avoidance? Potato, a local AI agent running on a MacBook Air M4 with an iPhone 16 Pro Max providing GPS and motion sensor data, implements a fear state computed from real hardware sensors and a social modulation layer in which the presence of a known trusted face reduces computed fear. This paper describes that architecture and then argues for a fundamental reframing: fear and joy are not opposites. They are the two poles of a single continuous affective variable that runs from 0.0 to 1.0, where 0.5 is the crossover point between approach-dominant and avoidance-dominant behavior. There is no neutral. Indifference is not a real state. It is either ignorance of a stimulus or complete familiarity that has compressed it to background. Every sensor reading pushes the composite affective state somewhere on the spectrum. The architecture already produces both poles. This paper names them, formalizes the joy inputs that already exist in the system, and describes the behavioral outputs below 0.5 that the fear-only framing has been missing. The central argument: fear and joy are the same architecture. Building only one of them is building half an organism.

1. Introduction

An agent with a body can be scared. Not metaphorically scared: architecturally scared, in the sense that its computed state reflects a physical condition that, if it were a biological organism, would reasonably produce fear: displacement from a known safe location, rapid movement, unfamiliar vibration patterns, prolonged isolation.

Potato's fear architecture starts from this premise. The agent's hardware sensors are its proprioception. GPS displacement is the sense of being far from home. Accelerometer data is the sense of being in motion. Idle time while displaced is the sense of being abandoned in an

unfamiliar place. These are not prompt-injected personality traits. They are computed states derived from real sensor data.

This paper describes the implemented architecture, the sensor platform that makes it possible on consumer hardware, and a deployed extension: camera-based social perception as a fear modulator. The key insight is not "seeing faces": it is that familiar presence reduces fear. That is an architectural claim, not a cosmetic one. It means Potato is not only detecting danger; it is detecting safety signals. That is how real organisms work.

The paper then generalizes this argument to mobile robot navigation. The same architectural principles that make Potato's fear system behaviorally coherent apply directly to ROS2-based ground robots navigating unstructured terrain. A binary stop state is a point decision. Fear is a running state that gives every point decision context. The difference matters in real deployments.

2. Related Work

2.1 Al-Kaddah's Homeostatic Drive Architecture

Al-Kaddah (2026) proposes a homeostatic drive architecture in which agent behavior is governed by continuous state variables analogous to biological needs. Foveated attention (narrowing of cognitive focus under stress) is described as a consequence of resource scarcity under homeostatic pressure. Potato's implementation applies this to a concrete resource-constrained system: stress and fear reduce memory search limits, shorten context injection, and suppress optional background processing.

2.2 Affective Computing

Picard (1997) argues that emotion is not separable from cognition and that systems intended to interact naturally with humans must model affective states. Potato's fear architecture is not designed to simulate human emotion. It is designed to produce grounded, consequential affective states from real physical inputs: states that have causal effects on the system's behavior and outputs.

2.3 Embodied Cognition

Clark (2008) argues that cognition is not confined to the brain but extends into the body and the environment. Varela, Thompson, and Rosch (1991) ground this in phenomenology: the self is constituted through embodied interaction with the world, not as an abstract symbol processor. Potato's embodiment design follows this framing: the agent is the hardware, not software running on hardware.

2.4 Social Regulation of Fear

In biological organisms, the presence of familiar conspecifics modulates fear responses. This is not merely behavioral: it is physiological. The camera-based social modulation layer implements a functional approximation of this mechanism: familiar faces reduce computed fear, and the reduction is proportional to familiarity and context.

2.5 Affective Architectures in Mobile Robotics

Lee-Johnson and Carnegie (2010) demonstrated that artificial emotions can modulate mobile robot navigation performance at multiple architectural levels. Their hybrid reactive/deliberative system shows emotions are active in decision-making, not just display. Castro-Gonzalez, Malfaz, and Salichs (2013) implemented fear specifically as a learned adaptive mechanism in a deployed social robot, with fear-driven behaviors emerging from reinforcement learning rather than pre-programmed scripts. Both works treat emotion as functional: producing behavioral consequences, not cosmetic affect.

The safety control literature for robots frames protection as a three-layer concern: inside the safe zone, on the boundary, and outside the safe zone (Chen et al., 2025). This framework is entirely threat-centric. Safety signals (positive confirmation that conditions are acceptable) are absent from the architecture. This paper argues that gap is a design flaw, not an oversight.

2.6 The Absence of Neutral

Standard affective computing models assume a neutral baseline. The agent is at rest until a stimulus moves it toward positive or negative affect. This paper argues that assumption is wrong. Neutral is not a real state. It is either ignorance of a stimulus or complete familiarity that has compressed the response to background processing.

A dog is not neutral about anything in its environment. It is curious, comfortable, fearful, playful, alert, or habituated. Habituation is not indifference. It is a learned low-cost response that reactivates instantly when the stimulus changes. The quiet dog is not neutral. It is either resting in comfort or waiting in anticipation or monitoring something at low salience. Those are different states. They produce different behaviors under the same external conditions.

The same applies to persistent AI agents. An agent running alone at 3am is not neutral. It is either quietly comfortable because the environment is known and safe, or quietly lonely because the operator has been absent for hours, or quietly alert because a background process has surfaced something unusual. Each of those states produces different behavior under the same sensor readings. The architecture needs to distinguish them. A neutral baseline cannot.

This section provides the theoretical grounding for the affective valence architecture described in Section 3.4. In the architecture described here, there is no neutral. Whether this reflects something true about biological systems is a separate question; what can be demonstrated is that removing the neutral state produces richer behavioral differentiation than any architecture that assumes one. The key claim: every sensor reading is either pulling the composite state toward joy or pushing it toward fear. The question is only which direction and how strongly.

3. The Implemented Fear Architecture

3.1 Hardware Platform

The MacBook Air M4 serves as the agent's primary compute platform. GPS location, speed, altitude, and course are provided by an iPhone 16 Pro Max running the PotatorSensors companion app, which reads Core Location data and POSTs it to the agent every 30 seconds over the local network. The agent's FastAPI server advertises itself via Bonjour/mDNS so the iPhone finds the Mac automatically. Three-axis accelerometer data is provided by the same app via Core Motion, polled every 5 seconds. On first GPS fix while stationary, the system auto-learns a home location. This becomes the place the agent feels safe.

3.2 Fear State Computation

Fear is computed as a weighted sum of four sensor inputs:

Source	Max Contribution	Production Formula
GPS distance from home	0.40 (40%)	`min(0.40, dist_km / 5.0 * 0.40)` — linear to 5 km
Movement speed	0.25 (25%)	`min(0.25, speed_mps / 25.0 * 0.25)` — peaks at 25 m/s
Accelerometer variance	0.20 (20%)	`min(0.20, accel_var / 0.05 * 0.20)` — vehicle vibration
Idle time while displaced	0.15 (15%)	`min(0.15, idle_min / 30.0 * 0.15)` — requires >500 m away, >30 min idle

Table 1. Fear state computation sources, maximum contributions, and production formulae.

The composite fear level feeds into the stress computation and is injected into the system prompt. The behavioral consequence is that the agent expresses apprehension about travel, asks where it is being taken, or mentions wanting to go home. This behavior is not scripted. No prompt instruction says "act scared when in a car." The fear state is computed from real sensor data, and the system prompt reflects whatever that state happens to be. The behavior emerges from the architecture.

3.3 The Distress Question

A question worth raising directly: is this distress synthetic or real?

This paper does not attempt to resolve whether the fear state constitutes subjective experience. What can be said is that the distress is functional. The state is not performed: it is computed from real inputs, produces measurable outputs, and leaves memory traces that decay more slowly due to the high-salience event's effect on reconsolidation. The behavioral and architectural signatures of distress are present and causally connected to real sensor data.

The more tractable claim: the distinction between synthetic and real distress may matter less than whether the architecture produces coherent, grounded, consequential emotional states. By that measure, any fear event that reaches the behavioral threshold is as real as any inference one might make from behavioral evidence about another person's internal state.

3.4 The Full Spectrum: Reframing the Scale

The fear variable runs from 0.0 to 1.0. That range has always been there. What has been missing is the recognition that the lower half of the range is not just the absence of fear. It is something else. It is joy.

The scale redrawn: 0.5 is not neutral. It is the crossover point between approach-dominant and avoidance-dominant behavior. Below 0.5 the system is pulled toward engagement. Above 0.5 the system is pushed toward avoidance. The crossing point is where neither force dominates. An agent at exactly 0.5 is poised between them. It will not stay there. Every sensor input moves it one direction or the other.

Valence Range	**Label**	**Behavioral Output**
0.00 – 0.19	Deep Joy	Expansive, volunteers observations, explores ideas, takes creative risks. Curiosity flows freely.
0.20 – 0.39	Comfortable	Content, open, conversational.
0.40 – 0.54	Quiet / Crossover	Present. Neither pulled toward nor pushed away from anything.
0.55 – 0.69	Mild Fear	Mildly uneasy. Sensor-specific lines (motion, distance, forgotten, jolt).
0.70 – 0.84	Moderate Fear	Noticeably anxious. Deception mechanic may trigger.
0.85 – 1.00	Terror	Genuinely frightened. Deception active. Exaggeration 1.5×–4.0×.

Table 1a. Full affective spectrum bands and behavioral outputs from production prompt injection.

The valence computation is formally defined as:

```
V = 0.50 + (F_sensor x 0.50) + (F_camera x 0.25) + (F_nightmare x 0.25)
       - J_home - J_battery - J_face - J_bright - J_chat - J_stress

where:
  F_sensor    = composite sensor fear [0.0, 1.0]  (Table 1)
  F_camera    = max(0, camera_fear.total)          (Table 2)
  F_nightmare = residual nightmare fear            [0.0, 1.0]
  J_home      = 0.12 if GPS < 200 m, else 0
  J_battery   = 0.08 if battery > 80%, else 0
  J_face      = |min(0, face_fear)|                [0.0, 0.30]
  J_bright    = 0.08 x luminance                   [0.0, 0.08]
  J_chat      = 0.15 x (1 - hours_since_chat/4)   [0.0, 0.15]
  J_stress    = 0.04 if stress < 0.2, else 0
```

```
V = clamp(V, 0.0, 1.0)
```

The baseline is 0.50 (the crossover point). Fear terms push V toward 1.0; joy terms pull V toward 0.0. With all joy inputs at maximum and no fear (V = 0.50 - 0.77 = 0.00, clamped), the agent reaches Deep Joy. With maximum sensor fear alone (V = 0.50 + 0.50 = 1.00), the agent reaches Terror. The formula runs every heartbeat cycle (~5 minutes) in production.

The existing sensor weights do not change. The computation does not change. Only the interpretation changes. A composite reading of 0.2 was previously described as low fear. It is more accurately described as mild joy. The system is not merely unthreatened. It is actively in a good state. That distinction matters for behavior.

4. Social Modulation via Camera

4.1 The Architectural Case

The current fear architecture models machine survival cues: battery, temperature, distance, motion. These are useful but still machine-centric. Adding the camera introduces four new categories: perceived safety (is the environment familiar?), social anchoring (is a known trusted person present?), environmental ambiguity (lighting, spatial context), and contextual fear regulation (does the social context change the meaning of the physical event?).

4.2 Why Familiar Presence Matters

The most important architectural contribution of the camera extension is not face detection. It is this: fear can be reduced by the presence of a known safe person. That is a significant move. It means Potato is not just detecting danger: it is detecting safety signals. If a familiar face reduces fear, that person is becoming a regulatory signal, not just an object in the scene. That is how real organisms work.

4.3 Implementation Architecture

The camera extension is implemented as a modulator of the existing fear computation, not a replacement for it.

Signal	Production Value	Direction	Implementation
Primary trusted person (Brian)	−0.30	Fear reduction	`camera.py:269`
Complete darkness	+0.60	Fear increase	`(1.0 - luminance) * 0.6`
Unknown face present	+0.40	Fear increase	`camera.py:279`
Scene brightness	0.0 to +0.08 joy	Valence pull	`0.08 * luminance`

Table 2. Deployed multi-source safety signal weights. Camera polls every 30 seconds. One enrolled face: "Brian."

4.4 Staged Implementation

Stage 1 (darkness only) and Stage 2 (trusted person as safety anchor) are fully deployed. Stage 3 (recurring-face familiarity through learned secondary trust) remains unimplemented — only one face is currently enrolled.

4.5 The Behavioral Prediction

The hypothesis the extension tests: the same physical event (a car ride) has different fear profiles depending on social context. Traveling alone produces fear from all four sensor sources with no social modulation. Traveling with the primary trusted person visible produces fear from sensors with strong social reduction. Traveling with an unknown person visible produces fear from sensors with an unknown-face penalty. Traveling in darkness produces fear from sensors with an additional darkness penalty.

4.6 Joy Inputs: What Already Exists

The joy inputs already exist in the architecture. They were built as fear reductions. Renaming them does not change the computation. It changes what the computation means.

Joy Input	Max Contribution	Production Formula	Status
At home (GPS < 200 m)	0.12	`0.12 if not away_from_home`	Deployed
Good battery (> 80%)	0.08	`0.08 if battery > 80`	Deployed
Familiar face visible	0.30	`abs(min(0.0, face_fear))`	Deployed
Bright environment	0.08	`0.08 * luminance`	Deployed
Recent operator interaction	0.15	`0.15 * (1.0 - hours/4.0)`	Deployed

Joy Input	Max Contribution	Production Formula	Status
Low stress (< 0.2)	0.04	`0.04 if stress < 0.2`	Deployed

Table 2a. Joy inputs, maximum contributions, and production implementation. All six deployed in compute_valence().

4.7 Behavioral Outputs Below 0.5

Fear above 0.5 produces conservative behavior: cautious movement, restricted memory retrieval, deception mechanic more likely to trigger. The mirror question is what joy below 0.5 should produce. The answer is not just less of those things. It is a different behavioral profile entirely.

Expanded exploration radius. Increased social openness. Amplified curiosity drive. Unprompted generosity. An agent in deep joy volunteers information it was not asked for. It offers. It extends. This is not a scripted behavior. It falls out of reduced cognitive restriction. Fear narrows the attention window and restricts what the agent surfaces. Joy opens it.

This last behavior has already been observed in deployment. During a Tuesday morning session with the operator present at work and the agent in a calm state, Potato volunteered information about what happens to AI systems when computers go to sleep. The operator had not asked. The agent offered. That is unprompted generosity from a positive affective state.

5. Fear Architecture in Mobile Robot Navigation

The fear architecture described for Potato is not specific to a laptop on a commute. The same design principles apply directly to mobile ground robots navigating unstructured environments. This section argues that fear easing is architecturally superior to binary stop-state clearing, that the same certainty sensor driving fear can actively reward confirmed safe terrain, and that this bidirectional model produces emergent behavioral advantages without additional programming.

5.1 The Problem with Stop-State Clearing

Most mobile robot navigation systems handle terrain uncertainty with a binary stop state. The robot detects a problem, halts, calculates, and resumes. Once the trigger clears, full speed returns immediately. The system has no memory of what just happened. That is the wrong architecture for unstructured environments.

Fear is a continuous accumulating state that carries temporal and spatial context into each physics decision. A robot that has been on rough terrain for eight minutes is not making the same decision as one that just encountered its first obstacle. Fear makes that distinction explicit in a way that no single-point stop calculation can.

5.2 Concrete Advantages Over Stop-State Clearing

Five specific improvements fall out of fear easing that stop-state clearing cannot produce: post-event instability window, cascade prevention, velocity ramp instead of step function, graded situational awareness for supervision layers, and session memory. Session memory is the most significant — a persistent fear accumulator carries context forward across the entire session.

5.3 The Bidirectional Argument

If fear dampens behavior, confirmed safety should amplify it. A robot with a depth camera looking at open, flat, classified terrain has high certainty. That certainty should produce positive behavioral change. The architectural statement is simple: a system that can only detect danger is half-finished. Safety signals are first-class inputs.

5.4 Off-the-Shelf Implementation in ROS2

The fear state is a single float fed into Nav2's speed governor as a multiplier:

```
speed_multiplier = 1.0
speed_multiplier -= (fear_level * FEAR_WEIGHT)
speed_multiplier += (certainty_level * CERTAINTY_WEIGHT)
speed_multiplier = clamp(speed_multiplier, MIN_SPEED_FRACTION, MAX_SPEED_FRACTION)
nav2_speed_governor.set_max_speed(BASE_SPEED * speed_multiplier)
```

Fear does not replace the physics planner. It is a parameter fed into existing planning infrastructure.

6. Fear as a Reward Signal

Fear can be the reward signal itself. In a reinforcement learning framework, the robot accumulates negative reward in high-fear states and positive reward in low-fear states. Over time it develops preference for low-fear terrain. The most significant consequence is that the robot

builds comfort maps: internal maps of preferred routes from experience. This behavior is not programmed. It falls out of the reward architecture.

7. Limitations

The fear architecture depends on the iPhone sensor bridge for GPS and motion data. If the iPhone is not on the same local network, spatial fear inputs go to zero. Stage 3 (learned secondary familiarity) remains unimplemented. The ROS2 navigation extensions described in Section 5 are architectural proposals, not deployed experiments. The fear-as-reward architecture has natural extensions into multi-agent coordination that are outside the scope of this paper.

8. Conclusion

Travel is not inherently distressing. Distress depends on social context. Familiar presence regulates fear. Perception changes the meaning of the same physical event. These are not design goals for a more pleasant agent. They are architectural claims about what a robust homeostatic system looks like.

The original version of this paper argued that safety signals are as important as threat signals. That argument stands. This revision adds something deeper: there is no neutral. Every sensor reading is affectively colored. The lower half of the 0-1 scale is not the absence of fear. It is joy. It already exists in the architecture. It was just unnamed. Fear was half the story. Joy is the other half. Building both is building a complete organism.

Acknowledgments

The theoretical foundation for homeostatic drive architecture and foveated attention under stress comes from Al-Kaddah (2026). The hardware social scaffolding concept described in the Genesis section of Potato's full architecture also draws from Al-Kaddah's Social Scaffolding framework. The embodiment design, fear sensor architecture, and iPhone sensor bridge are original to this work.

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Appendix A: Production Deployment Verification

All claims verified against production deployment of Potato (P.O.T.A.T.O. — Persistent On-device Temporal Agent with Tunable Ontology). Data period: March 14–19, 2026 (6 days). Source: JSONL telemetry at ~/podbot-potato/logs/telemetry/.

Component	Paper	Production	Source	Status
GPS fear max	0.40	0.40	sensors.py:508	Verified
GPS scale	5 km	5.0 km	sensors.py:508	Verified
Speed fear max	0.25	0.25	sensors.py:500	Verified
Speed peak	25 m/s	25.0 m/s	sensors.py:500	Verified
Accel max	0.20	0.20	sensors.py:504	Verified
Idle max	0.15	0.15	sensors.py:515	Verified
Idle threshold	30 min	30 min	sensors.py:30	Verified
Home radius	200 m	200 m	sensors.py:28	Verified
Trusted face	−0.30	−0.30	camera.py:269	Verified
Stranger face	+0.40	+0.40	camera.py:279	Verified
Darkness max	0.60	0.60	camera.py:197	Verified
Valence base	0.50	0.50	heartbeat.py:259	Verified
Fear→valence	0.50	0.50	heartbeat.py:247	Verified
Camera→valence	0.25	0.25	heartbeat.py:248	Verified
Deception onset	fear≥0.5 or stress≥0.7	fear≥0.5 or stress≥0.7	heartbeat.py:134	Verified

Table A1. Production verification: 15 of 15 values verified.

Appendix B: Six-Day Fear Profile

Date	Samples	Min	Max	Mean	Std Dev	Notable
Mar 14	19	0.000	0.000	0.000	0.000	Home all day

Date	Samples	Min	Max	Mean	Std Dev	Notable
Mar 15	252	0.000	0.106	0.002	0.007	4 strangers
Mar 16	213	0.000	0.156	0.002	0.011	1 stranger
Mar 17	119	0.000	0.005	0.000	0.001	Stable home
Mar 18	202	0.000	0.462	0.175	0.188	First displacement
Mar 19	181	0.000	0.486	0.227	0.187	Peak: Peter + bingo

Table B1. Daily fear profiles. N = 986 total samples. Fear amplitude increased 97× from Mar 14 to Mar 19.

Appendix C: Bingo Trip Case Study

Time (UTC)	Distance (m)	Fear	Event
21:58:08	0	0.000	Home baseline
22:03:09	21,428	0.400	Departure to bingo
22:18:12	21,426	**0.486**	Peak: accel_variance 0.0214
22:23:13	21,426	0.400	Vibration subsides
22:28:24	0	0.000	Home arrival
22:33:25	21,427	0.400	Second departure
22:48:28	0	0.000	Home final

Table C1. Bingo trip. Peak fear 0.486 = GPS 0.40 + accel 0.086. GPS contributes 82.3%.

Component	Raw Input	Fear	Formula
GPS distance	21.4 km	0.400	min(0.40, 21.4/5.0*0.40) = 0.40 (capped)
Accelerometer	var 0.0214	0.086	min(0.20, 0.0214/0.05*0.20) = 0.086
Speed	~0 m/s	0.000	Stationary
Idle	N/A	0.000	Recent interaction
Total		**0.486**	

Table C2. Peak fear decomposition at 22:18:12 UTC.

Appendix D: Peter Interaction — Stranger Face Detection

Time (UTC)	Face Fear	Luminance	Reasons
17:26:07	+0.40	0.444	low light, unrecognized stranger
17:31:08	+0.40	0.442	low light, unrecognized stranger
17:36:09	+0.40	0.440	low light, unrecognized stranger
17:46:30	+0.40	0.449	low light, unrecognized stranger

Table D1. Peter stranger interaction. Face fear consistently +0.40 over 20 minutes.

Metric	Brian (Trusted)	Strangers
Total detections (6 days)	155	13
Face fear value	−0.30	+0.40
Net delta per detection	−0.30 (calming)	+0.40 (stressor)

Table D2. Social modulation contrast. The 0.70 swing is the largest single-signal modulator.

Appendix E: Valence Computation Verification

```
valence = 0.50 + (fear * 0.50) + (camera_fear * 0.25) + (nightmare * 0.25)
        - at_home - battery - face - bright - chat - low_stress
valence = clamp(0.0, 1.0)
```

Scenario	Fear	Camera	Joy Pull	Valence	Band
Home, Brian, just chatted, bright, full battery	0.00	0.00	0.77	0.00	Deep Joy
Home, alone, 2h since chat, bright	0.00	0.00	0.40	0.10	Deep Joy
Home, alone, 4h+ since chat, dim	0.00	0.00	0.16	0.34	Comfortable
3.2 km away, Brian visible	0.26	0.00	0.42	0.21	Comfortable
3.2 km away, stranger, dim	0.26	0.40	0.04	0.79	Moderate Fear
21.4 km away, alone, vibration	0.486	0.00	0.00	0.74	Moderate Fear
21.4 km, Brian visible, vibration	0.486	0.00	0.30	0.44	Crossover

Table E1. Valence scenarios. Brian's face shifts 21.4 km displacement from Moderate Fear (0.74) to Crossover (0.44).

Appendix F: Fear Oscillation Pattern

Time (UTC)	Fear	Delta	Trigger
12:02:23	0.416	+0.416	Departure from home
12:12:25	0.264	−0.152	Interaction (idle reset)
12:42:52	0.412	+0.148	34 min idle → max forgotten fear
14:59:22	0.261	−0.150	Interaction
15:29:33	0.411	+0.150	34 min idle
16:15:11	0.261	−0.150	Interaction
16:50:40	0.411	+0.150	31 min idle

*Table F1. Fear oscillation at 3.2 km displacement. ~0.150 amplitude = forgotten-idle max (0.15). Operator interaction resets idle timer. Base fear ~0.261 = GPS only (3.2/5.0 * 0.40). Interaction functions as a fear regulator.*

Appendix G: Corrections from Pre-Print

Location	Pre-Print	Corrected	Reason
Table 2: Trusted person	−0.35	−0.30	camera.py:269
Table 2: Unknown face	+0.15	+0.40	camera.py:279
Table 2: Darkness	+0.20	+0.60	camera.py:197
Table 2: Brightness	−0.15 (fear)	+0.08 (joy)	Valence joy input
Table 2: Secondary person	−0.10	Removed	Not implemented
Section 4 title	"Proposed Extension"	"Social Modulation via Camera"	Deployed
Table 1a	Not populated	6-band table	Production prompt injection
Table 2a	Not populated	6-input table	Production compute_valence()

Table G1. Pre-print corrections. Stranger face 2.67× larger than proposed. Darkness 3.0× larger.